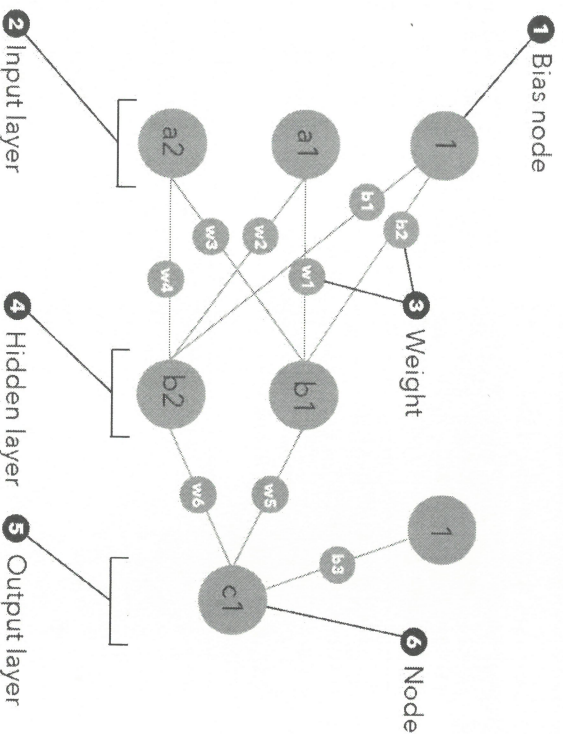


Pre-Stage: Creating the Network Structure

In the previous chapters, we touched on a variety of neural network architectures such as CNNs and RNNs. Although each is slightly different, all have a structure - shell - that is made up of similar parts. These parts are called *hyperparameters*, and include elements such as the number of layers, nodes and the learning rate.

Hyperparameters are essentially fine tuning knobs that can be tweaked to help a network successfully train. In fact, they are determined before a network trains, and can only be adjusted manually by the individual or team who created the network. The network itself does not adjust them.

You can see some of these hyperparameters in the feedforward network below:



In this chapter we are going to take a brief pause to explore these hyperparameters because they are extremely important. Not only are they tuning knobs that help a network train, but all of a network's mathematical calculations depend on them being in place. We will begin by defining the two categories of hyperparameters, and then briefly explore each.

Types of Hyperparameters

Broadly speaking there are two categories of hyperparameters: required and optional.

Required Hyperparameters Include:

- Total number of input nodes
- Total number of hidden layers
- Total number of hidden nodes in each hidden layer
- Total number of output nodes
- Weight values
- Bias values
- Learning Rate

Optional Hyperparameters Include:*

- Learning rate schedule (a.k.a learning rate decay)
- Momentum
- Mini-batch size
- Weight decay
- Dropout

*This list is not exhaustive !

Now, let's take a brief look at each of the required hyperparameters. Please note that many new terms will be introduced below, and we will explain each as we move throughout the stages.

Input Node

An input node contains the *input* of the network and this input is *always* numerical. If the input is not numerical by default, it is always converted. For example: